



Problem Solving With C
UE24CS151B

By,
Prof. Bivas Bhattacharya,
Assistant Professor

1. **(1 Mark):** What is the primary function of GCC?
2. **(1 Mark):** How do you specify a single-line comment in C?
 -
3. **(2 Marks):** Briefly explain the difference between a compile-time error and a run-time error.
4. **(2 Marks):** What is the purpose of the `-o` flag when using GCC, and provide an example of its use?
5. **(3 Marks):** Describe the purpose of the linking step in the C compilation process, and why it's sometimes necessary to link against libraries.
6. **(3 Marks):** Explain the importance of following C compiler standards like C99 or C11.
7. **(4 Marks):** Outline the steps involved in compiling and running a C program using the separate compilation and linking method (Way 2 from the document). Include the GCC commands used at each stage.
8. **(5 Marks):** Discuss the different phases of the Program Development Life Cycle (PDLC). Choose three phases and describe the activities typically performed in each.
9. **(5 Marks):** Explain the different types of errors that can occur during program development (compile-time, link-time, run-time, logical). Provide an example of each type of error and describe how it might be detected.
10. **(10 Marks):** You have a project with three C files: `main.c`, `functions.c`, and `utilities.c`. `functions.c` uses functions from a math library (`libm`). Write the GCC commands to:
 - Compile each file separately into object files. (3 marks)
 - Link the object files together, including the math library, to create an executable named "myprogram". (3 marks)
 - Explain how the compilation process can be streamlined using a "make" utility for larger projects. (3 marks)

